Omer Abdur Rehman

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Profile

Computer Systems Engineering graduate from The Islamia University of Bahawalpur with hands on experience building and deploying real world software. Developed a production grade AI driven stock prediction system as a Final Year Project integrating machine learning, NLP sentiment analysis, and a cross-platform mobile app using Python, and Flutter. Comfortable across the full stack: Python, C++, data analytics, and app development, with a growing focus on applied AI, intelligent systems, and Machine Learning. Brings an engineering mindset, clean implementation habits, and the practical experience of taking a system from idea to deployment.

Experience

Punjab Information Technology Board (PITB) - Flutter App Development (June 2025 – August 2025)
I completed a two-month Flutter internship at the Punjab Information Technology Board (PITB), developing cross-platform mobile features using Flutter. Implemented frontend and backend integration, improved UI responsiveness, and optimized application performance in a structured development environment.

Punjab Information Technology Board (PITB) - AI/ML Intern (August 2026 – September)
I completed an AI/ML internship at the Punjab Information Technology Board (PITB), working on real-world machine learning and artificial intelligence projects. Developing and evaluating machine learning models, performing data preprocessing and analysis, and implementing computer vision solutions using Python and deep learning frameworks in a structured development environment.

PROJECTS

Smart Stock Predictor: A Next Generation AI Platform for Predictive Financial Intelligence

- Designed and developed, as part of my Final Year Project, an AI driven stock prediction system using Random Forest regression with 19 technical indicators (RSI, MACD, ATR, ADX).
- Built a FastAPI backend with time series validation and lag feature engineering to reduce data leakage risks.
- Implemented FinBERT and VADER sentiment analysis pipeline for financial news processing using Yahoo Finance data.
- Developed a Flutter based mobile application with real time dashboards across 17 screens for portfolio visualization.
- Deployed the system on Render with Firebase authentication and a scalable ML pipeline architecture.

Advanced Greenhouse Controller System

- Engineered an AT89C51 based greenhouse monitoring system for real time temperature and humidity tracking.
- Integrated ADC0804 sensor interfacing with a 16×2 LCD and LED indicators for live environmental status display.
- Designed a cost-efficient embedded system demonstrating hardware level sensor integration and control logic.

Intelligent Traffic Light Control System

- Developed a dual Arduino traffic control system using I2C communication for 4-way intersection management.
- Implemented LDR based automatic night mode and adaptive timing using potentiometer control.
- Designed a non blocking finite state machine for real time traffic signal responsiveness.

Water Level Monitoring and Alert System

- Built an automated water level monitoring system with threshold-based pump control to prevent overflow and dry running.
- Implemented real time visual and buzzer based alert system for water level thresholds.
- Designed efficient timing logic for continuous monitoring without system blocking.

Fire Alert System Utilizing the AT89C51 Microcontroller

- Developed a real time fire detection system using AT89C51 microcontroller with LCD status output.
- Implemented buzzer-based emergency alert system for immediate hazard notification.
- Engineered a safety critical embedded solution with reliable sensor-based detection.

Waste Detection — YOLOv8

- Developed an object-detection pipeline for identifying common waste categories.
- Trained YOLOv8 on a custom/combined dataset using Google Colab GPU.
- Achieved 0.856 mAP@0.5 and 0.665 mAP@0.5:0.95.
- Performed class-level error analysis and evaluated model predictions on unseen images.

Portfolio-Pilot — AI Portfolio Automation Agent

- Developed a modular AI agent to transform technical project details into professional portfolio, GitHub, and LinkedIn content.
- Built tools for project extraction, README generation, LinkedIn post creation, learning roadmaps, and automated content review.
- Integrated Ollama-based local LLM inference for private and cost-efficient content generation.
- Designed the workflow in Python and Google Colab to process AI/ML and computer vision projects into structured, recruiter-friendly documentation.

SKILLS

- Python
- C++
- Data Analytics
- Artificial Intelligence
- Machine Learning
- App Development (Flutter)

EDUCATION

• **BSc Computer System Engineering** (2022 – 2026)
The Islamia University of Bahawalpur

• **FSc pre engineering** (2019 – 2021)
Punjab Group of Colleges, Bahawalpur

• **Matric** (2017 – 2019)
Stars High School, Bahawalpur

EXTRA CURRICULAR ACTIVITIES

- Joint Secretary I, Innovation and Commercialization Society (2025–2026)
- Joint Secretary, Engineering Student Society (2025–2026)
- General Secretary, Departmental Club of CSE (2023–2024)